Student Assignment Brief

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Contents

Assignment Information

Module Name:	Intelligent Information Retrieval
Module Code:	ST7071CEM
Assignment Title:	Individual Coursework / Viva
Assignment Due:	23rd January, 2026

Assignment Credit:	15 credits
Word Count:	3000 words
Assignment Type:	Coursework
Grading:	75%

Assessment Overview

You will be provided with an overall grade between 0% and 100%. You have one opportunity to pass the assignment at or above 40%.

Important Notice

The work you submit for this assignment must be your **own independent work**. More information is available in the 'Assignment Task' section of this assignment brief.

Assessed Module Learning Outcomes

The Learning Outcomes for this module align with the marking criteria which can be found at the end of this brief. Ensure you understand the marking criteria to ensure successful achievement of the assessment task.

1. Demonstrate a sound knowledge of information retrieval principles.
2. Apply main data structures used in index construction in Python or a similar high-level language.
3. Implement a typical web crawler and query processor, in Python or similar high-level language.
4. Acquire knowledge and skill to apply common machine learning methods for text classification and document clustering.
5. Build the outline of a minimum viable vertical search engine for text retrieval.

Assignment Task

There are two tasks in this coursework. You can use any general-purpose programming language of your choice to perform these tasks. However, Python is recommended. The tasks are specified next.

Task 1: Search Engine

Create a vertical search engine comparable to Google Scholar, but specialized in retrieving just papers/books published by a member of Coventry University's Research Centre for Computational Science and Mathematical Modelling : https://pureportal.coventry.ac.uk/en/organisations/ics-research-centre-for-computational-science-and-mathematical-mo

That is, at least one of the co-authors is a member of this department.

Your system crawls the relevant web pages and retrieves information about all available publications. For each publication, it extracts available data (such as authors, publication year, and title) and the links to both the publication page and the author's profile (also called "pureportal" profile) page.

Make sure you that your crawler is polite, i.e., it preserves the robots.txt rules and does not hit the servers unnecessarily or too fast.

Because of low rate of changes to this information, your crawler may be scheduled to look for new information, say, once per week, but it should ideally be able to do so automatically, as a scheduled task. Every time it runs, it should update the index with the new data. Make sure you apply the required pre-processing tasks to both the crawled data and the users' queries.

From the user's perspective, your system provides an interface like Google Scholar's main page, where the user may enter in queries/keywords concerning the resources they wish to locate. The results will then be shown by your system, arranged by relevancy, in a manner like Google Scholar. However, the search results are limited to this department members' publications. Unless you aim to earn a score of 70 or above, the user interface does not need to be web-based (like Google Scholar), and the usual Python interface in your IDE would do. However, for more usability, it would be preferable to be able to click on the printed links rather than copying and pasting them into a browser.

Task 2: Document Clustering

First, collect several documents that belong to different categories, namely Business, Entertainment and Health. Each document should be at least one sentence (the longer is usually the better). The total number of documents is up to you but should be at least 100 (the more is usually the better). You may collect these documents from publicly available web sites such as BBC news websites, but make sure you preserve their copyrights and terms of use and clearly cite them in your work. You may simply copy-paste such texts manually and writing an RSS feed reader/crawler to do it automatically is NOT mandatory. Once you have collected sufficient documents, cluster them using a standard clustering method (e.g., K-means).

Finally, use the created model to assign a new document to one of the existing clusters. That is, the user enters a document (e.g., a sentence) and your system outputs the right cluster.

NOTE: You must show in your report and viva that your system suggests the right cluster for variety of inputs, e.g., short and long inputs, those with and without stop worlds, inputs of different topics, as well as more challenging inputs to show the system is robust enough.

Submission Instructions

Requirement	Details
File Naming	NAME_studentID
File Format	.docx/.pdf format
Submission Method	Campus 4.0 platform (submission link provided 2 weeks before deadline)

Marking and Feedback

How will my assignment be marked?

Your assignment will be marked by the Module Team using standardized criteria.

How will I receive grades and feedback?

Provisional marks will be released once internally moderated. Feedback will be provided alongside grades release within 2 weeks (10 working days).

What will I be marked against?

Details of the marking criteria for this task can be found in the Assessment Marking Criteria section at the end of this brief.

Grade Requirements

You must achieve 40% or above to pass this assessment. Ensure you understand the marking criteria for successful completion.

Assignment Support and Academic Integrity

Getting Help

If you have any questions about this assignment, please meet your respective module leader or teacher for more information.

Language Standards

You are expected to use effective, accurate, and appropriate language within this assessment task.

Academic Integrity

The work you submit must be your own. All sources of information need to be acknowledged and attributed; therefore, you must provide references for all sources of information and acknowledge any tools used in the production of your work, **excluding Artificial Intelligence (AI)**.

We use detection software and make routine checks for evidence of academic misconduct. Definitions of academic misconduct, including plagiarism, self-plagiarism, and collusion can be found in Student handbook in Campus 4.0.

All cases of suspected academic misconduct are referred to for investigation, the outcomes of which can have profound consequences to your studies.

Support for Students with Disabilities

If you have a disability, long-term health condition, specific learning difference, mental health diagnosis or symptoms, contact the Student Support Office for assistance.

Unable to Submit on Time?

If events prevent you from submitting on time, guidance on extenuating circumstances is available in the Student Handbook or from the Student Support Office.

Administration of Assessment

Module Leader Name:	Mr. Siddhartha Neupane
Module Leader Email:	Stw0084@softwarica.edu.np
Assignment Category:	Written
Attempt Type:	Standard
Component Code:	Coursework

Assessment Criteria

Marking Requirement for each Task

Requirements	**Marks**
Fully working crawler component *Must completely crawl the CU profile in google scholar and subsequent links in BFS manner. It should be scheduled to re-crawl to extract new data automatically.*	25
Construction of Inverted Index *Construction of the index based on appropriate data structures studied in the module as opposed to naïve database tables. The index should be updated once new data are received from the crawler component.*	15
Fully working query processor component *Displaying results relevant to given queries*	25
Fully working document classification component *A standard classification algorithm such as Naïve Bayes Classifier must be used. In addition, the learned model must be used to identify the right classification (e.g., Business, Health or Politics) for a given input. Various inputs must be used both in the report (via screenshots) and the viva to show that the system is accurate and robust.*	25
Overall usability *Acceptable response time, accuracy of results, nice interface, readable results sorted by relevance, subject grouping, and anything else that might affect the usability of the product.*	10

Markin allocation guideline to students

0-39	40-49	50-59	60-70	70+	80+
Work mainly incomplete and /or weaknesses in most areas	Most elements completed; weaknesses outweigh strengths	Most elements are strong, minor weaknesses	Strengths in all elements	Most work exceeds the standard expected	All work substantially exceeds the standard expected

Grade	Answer Relevance	Report	Code And Results
First ≥70	Innovative response, answers the question fully, addressing the learning objectives of the assessment task. Evidence of critical analysis, synthesis and evaluation.	A clear, consistent in-depth critical and evaluative report. Engagement with theoretical and conceptual analysis. Correctly referenced.	Code is well written and follows a logical structure. Fully Working Crawler Components and Construction of Inverted Index.
Upper Second 60-69	A very good attempt to address the objectives of the assessment task with an emphasis on those elements requiring critical review.	A generally clear line of critical and evaluative argument is presented. Relationships between statements and sections are easy to follow, and there is a sound, coherent structure. Correctly referenced in the main.	Code is readable and functions as expected. Crawler and Inverted index component should be working
Lower Second 50-59	Competently addresses objectives, but may contain errors or omissions and critical discussion of issues may be superficial or limited in places.	Some critical discussion, but the argument is not always convincing, and the work shows only a partial understanding of the key concepts. Referencing is not always correctly presented.	Code functions as expected. Crawler components works properly and partial construction of Inverted Index
Third 40-49	Addresses most objectives of the assessment task, with some notable omissions. The structure is unclear in parts, and there is limited analysis.	Limited understanding of the theoretical concepts. Limited justification of method and results. Referencing has some errors.	Code has most of the functionality implemented. Limited construction of inverted index and crawler component works partially.
Fail <40	Some deviation from the objectives of the assessment task. May not consistently address the assignment brief. At the lower end fails to answer the question set or address the learning outcomes. There is minimal evidence of analysis or evaluation.	Descriptive with no evidence of theoretical engagement. At the lower end displays a minimal level of understanding. Poor presentation of references.	Code has some functionality implemented. With no implementation of construction of Inverted Index, Crawler component not Working.
Late Submission	0	0	0